INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Nasarawa State, Nigeria

ACICP 101 — Physics & Electrical Circuit Fundamentals for ICS/SCADA

Week 1 Assignment 01: Circuit-Calculation Worksheet

Lab Title:	Circuit-Calculation Worksheet (Ohm's Law, Power, Series/Parallel, 4-20 mA Scaling)
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Executive Summary

This worksheet solves the Week 1 Circuit-Calculation Assignment for ACICP 101. It covers basic Ohm's Law and power calculations, series and parallel resistor networks, 4-20 mA instrumentation scaling used in ICS/SCADA systems, and a troubleshooting calculation on a relay coil and a voltage divider. Every question is solved using the simplest classroom method: state the formula, substitute the given values, then write the final answer with its correct unit, exactly as taught on the board. All workings are shown in full as required by the assignment instructions.

Lab Objectives

To apply Ohm's Law (V = I x R) to calculate current, voltage, resistance and power in simple DC circuits.

To calculate total resistance, current and power in series, parallel and mixed resistor networks.

To apply 4-20 mA current loop scaling used in ICS/SCADA instrumentation.

To calculate cable resistance and voltage drop in a field instrumentation loop.

To use calculation to troubleshoot a faulty relay coil and design a voltage divider.

Tools and Resources Used

Scientific calculator - used to carry out arithmetic and check working.

Formula sheet provided in the assignment (Ohm's Law, Power, Series, Parallel, Voltage Divider, 4-20 mA Scaling).

Pen and paper method (chalkboard style) - formula, substitution, answer.

Methodology and Solutions

Every question below is solved in three simple steps, the same way a teacher solves it on the board: (1) write the formula, (2) substitute the given numbers into the formula, (3) write the final answer with its unit. Units are written plainly (V, A, mA, ohm, kOhm, W, mW) as would be written with chalk on a board.

Part A: Basic Electrical Quantity Calculations

Question A1

A 24 V DC supply is connected to a 160 ohm load. Find current and power.

Given: V = 24 V, R = 160 ohm

Formula: I = V / R

Substitute: I = 24 / 160

I = 0.15 A = 150 mA

Formula: P = V x I

Substitute: P = 24 x 0.15

Answer: P = 3.6 W

Question A2

A PLC input module draws 8 mA from a 24 V supply. Find resistance and power.

Given: I = 8 mA = 0.008 A, V = 24 V

Formula: R = V / I

Substitute: R = 24 / 0.008

Answer: R = 3000 ohm = 3 kOh

Formula: P = V x I

Substitute: P = 24 x 0.008

Answer: P = 0.192 W

Question A3

A 12 V indicator lamp is rated at 3 W. Find normal current and resistance.

Given: V = 12 V, P = 3 W

Formula: I = P / V

Substitute: I = 3 / 12

Answer: I = 0.25 A = 250 mA

Formula: R = V / I

Substitute: R = 12 / 0.25

Answer: R = 48 ohm

Question A4

A resistor carries 75 mA and dissipates 0.45 W. Find voltage and resistance.

Given: I = 75 mA = 0.075 A, P = 0.45 W

Formula: V = P / I

Substitute: V = 0.45 / 0.075

Answer: V = 6 V

Formula: R = V / I

Substitute: R = 6 / 0.075

Answer: R = 80 ohm

Question A5

A 0.25 W resistor is connected across 24 V. Find the minimum resistance so it is not overloaded.

Given: P (max) = 0.25 W, V = 24 V

Formula: R = V^2 / P

Substitute: R = 24^2 / 0.25 = 576 / 0.25

Answer: R(minimum) = 2304 ohm

Note: any resistance below 2304 ohm will let the power exceed 0.25 W, so the resistor must be 2304 ohm or higher.

Part B: Series, Parallel and Mixed Networks

Question B1

120 ohm, 330 ohm and 470 ohm resistors in series across 24 V DC.

Given: R1 = 120 ohm, R2 = 330 ohm, R3 = 470 ohm, V = 24 V

Formula: R(total) = R1 + R2 + R3

Substitute: R(total) = 120 + 330 + 470

Answer: R(total) = 920 ohm

Formula: I = V / R(total)

Substitute: I = 24 / 920

Answer: I = 0.0261 A = 26.1 mA (same current flows through all three, series circuit)

Formula: V = I x R (for each resistor)

V1 = 0.0261 x 120 = 3.13 V

V2 = 0.0261 x 330 = 8.61 V

V3 = 0.0261 x 470 = 12.26 V

Check: V1 + V2 + V3 = 3.13 + 8.61 + 12.26 = 24.0 V (equals supply, correct)

Formula: P = V x I (for each resistor)

P1 = 3.13 x 0.0261 = 0.082 W

P2 = 8.61 x 0.0261 = 0.225 W

P3 = 12.26 x 0.0261 = 0.320 W

Question B2

240 ohm and 360 ohm loads in parallel across a 24 V supply.

Given: R1 = 240 ohm, R2 = 360 ohm, V = 24 V

Formula: 1/R(total) = 1/R1 + 1/R2

Substitute: 1/R(total) = 1/240 + 1/360 = 0.004167 + 0.002778 = 0.006944

Answer: R(total) = 1 / 0.006944 = 144 ohm

Formula: I = V / R (each branch, since voltage is the same across both in parallel)

I1 = 24 / 240 = 0.1 A = 100 mA

I2 = 24 / 360 = 0.0667 A = 66.7 mA

Formula: I(supply) = I1 + I2

Answer: I(supply) = 100 + 66.7 = 166.7 mA

Formula: P(total) = V x I(supply)

Substitute: P(total) = 24 x 0.1667

Answer: P(total) = 4.0 W

Question B3

100 ohm resistor in series with a parallel network of 220 ohm and 330 ohm. Supply = 24 V DC.

Step 1: Find resistance of the parallel section (220 ohm and 330 ohm)

Formula: 1/Rp = 1/R2 + 1/R3

Substitute: 1/Rp = 1/220 + 1/330 = 0.004545 + 0.003030 = 0.007576

Rp = 1 / 0.007576 = 132 ohm

Step 2: Find total resistance

Formula: R(total) = R1 + Rp

Substitute: R(total) = 100 + 132

Answer: R(total) = 232 ohm

Step 3: Find total current

Formula: I(total) = V / R(total)

Substitute: I(total) = 24 / 232

Answer: I(total) = 0.1034 A = 103.4 mA

Step 4: Find voltage across the parallel section

Formula: V(parallel) = I(total) x Rp

Substitute: V(parallel) = 0.1034 x 132

Answer: V(parallel) = 13.66 V

Step 5: Find the current through each branch of the parallel section

I(220 ohm) = V(parallel) / 220 = 13.66 / 220 = 0.0621 A = 62.1 mA

I(330 ohm) = V(parallel) / 330 = 13.66 / 330 = 0.0414 A = 41.4 mA

Check: 62.1 mA + 41.4 mA = 103.5 mA (matches total current, small rounding)

Part C: Instrumentation and ICS/SCADA Signal Calculations

Question C1

A 4-20 mA pressure transmitter is scaled for 0-10 bar. Find the pressure at 4 mA, 8 mA, 13.6 mA, 16 mA and 20 mA.

Formula: Percent = (I - 4) / 16 x 100%, then Pressure = Percent x 10 bar / 100

(this is the same as: Pressure = (I - 4) / 16 x 10 bar)

At 4 mA: P = (4 - 4)/16 x 10 = 0 bar

At 8 mA: P = (8 - 4)/16 x 10 = (4/16) x 10 = 2.5 bar

At 13.6 mA: P = (13.6 - 4)/16 x 10 = (9.6/16) x 10 = 6 bar

At 16 mA: P = (16 - 4)/16 x 10 = (12/16) x 10 = 7.5 bar

At 20 mA: P = (20 - 4)/16 x 10 = (16/16) x 10 = 10 bar

Question C2

A 250 ohm burden resistor converts a 4-20 mA signal to voltage. Find voltage at 4 mA, 12 mA and 20 mA.

Formula: V = I x R

At 4 mA: V = 0.004 x 250 = 1 V

At 12 mA: V = 0.012 x 250 = 3 V

At 20 mA: V = 0.020 x 250 = 5 V

Question C3

Field cable has 0.055 ohm per metre per conductor. Transmitter is 180 m from the PLC. Find total loop cable resistance and voltage drop at 20 mA.

Note: a current loop needs two conductors (go and return), so total wire length = 2 x 180 m = 360 m

Formula: R(cable) = length x resistance per metre

Substitute: R(cable) = 360 x 0.055

Answer: R(cable) = 19.8 ohm

Formula: V(drop) = I x R(cable)

Substitute: V(drop) = 0.020 x 19.8

Answer: V(drop) = 0.396 V (approximately 0.40 V)

Question C4

PLC analogue input needs at least 11 V at the transmitter terminals. Supply = 24 V DC, burden = 250 ohm, loop current = 20 mA, cable resistance as above. Is the supply adequate?

Voltage used by burden resistor: V(burden) = I x R(burden) = 0.020 x 250 = 5 V

Voltage used by cable: V(cable) = 0.396 V (from C3 above)

Formula: V(at transmitter) = V(supply) - V(burden) - V(cable)

Substitute: V(at transmitter) = 24 - 5 - 0.396

Answer: V(at transmitter) = 18.6 V

Compare: 18.6 V is greater than the required 11 V.

Conclusion: Yes, the 24 V supply is adequate for this loop.

Part D: Troubleshooting Calculation (10 marks)

Question D1

A relay coil is labelled 24 V DC, 120 ohm. Find the expected current. Measured current is only 50 mA. Find the apparent resistance and state two likely causes.

Step 1: Expected current

Formula: I = V / R

Substitute: I = 24 / 120

Answer: I(expected) = 0.2 A = 200 mA

Step 2: Apparent resistance from the measured current

Given: I(measured) = 50 mA = 0.05 A

Formula: R(apparent) = V / I(measured)

Substitute: R(apparent) = 24 / 0.05

Answer: R(apparent) = 480 ohm

Two likely causes of this higher resistance (lower current than expected):

1. The coil winding is partially open or damaged (burnt/broken turns), which increases its resistance.

2. A loose, corroded or dirty terminal/connection in the coil circuit is adding extra resistance in series.

Question D2

A sensor input expects 5 V from a divider supplied by 24 V. R2 = 5 kOhm (from output to 0 V). Find R1 (from 24 V to output), the divider current, and power in each resistor.

Step 1: Find R1 using the voltage divider formula

Formula: V(out) = V(in) x R2 / (R1 + R2)

Substitute: 5 = 24 x 5000 / (R1 + 5000)

Cross multiply: 5 x (R1 + 5000) = 24 x 5000

5R1 + 25000 = 120000

5R1 = 95000

Answer: R1 = 19000 ohm = 19 kOhm

Step 2: Find divider current

Formula: I = V(in) / (R1 + R2)

Substitute: I = 24 / (19000 + 5000) = 24 / 24000

Answer: I = 0.001 A = 1 mA

Step 3: Find power in each resistor

Formula: P = I^2 x R

P(R1) = (0.001)^2 x 19000 = 0.019 W = 19 mW

P(R2) = (0.001)^2 x 5000 = 0.005 W = 5 mW

Check: P(total) = V x I = 24 x 0.001 = 0.024 W = 19 mW + 5 mW = 24 mW (correct)

Conclusion

All calculations in this worksheet were carried out using the basic Ohm's Law and power formulas, the series and parallel resistance rules, and the 4-20 mA scaling formula. Every answer was obtained step by step: formula first, then substitution, then final answer with its correct unit, the same simple method used on the chalkboard in class. This exercise reinforced how voltage, current, resistance and power relate to one another in simple DC circuits, in resistor networks, and in 4-20 mA instrumentation loops used in ICS/SCADA systems, and showed how calculation can be used to check whether a measured fault value (relay coil) matches the expected value.

Student Declaration

I confirm that this submission is my own work and that all sources, screenshots, simulations, and calculations used have been honestly presented.

I understand that unsafe electrical practice, copied work, or fabricated screenshots may attract disciplinary action.

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